Machine Intelligence 1

3.3 Bayesian Inference and Neural Networks

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3.3.1 Generative Models

Generative models

- observations: $\underline{\mathbf{z}}^{(\alpha)} = (\underline{\mathbf{x}}^{(\alpha)}, \underline{\mathbf{y}}_T^{(\alpha)})$ for $\alpha = 1, \ldots, p$

$$p_{(\underline{\mathbf{z}})} \quad = \quad p_{(\underline{\mathbf{y}}_T | \underline{\mathbf{x}})} \quad \cdot \quad p_{(\underline{\mathbf{x}})}$$

- most of our previous approaches:
  - $\rightsquigarrow$ construction of a parametrized class $y_{(\underline{\mathbf{x}};\underline{\mathbf{w}})}$ of (deterministic) predictors
  - $\rightsquigarrow$ inference is based on ONE selected (optimal) predictor $y_{(\underline{\mathbf{x}};\underline{\mathbf{w}}^*)}$

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generative model approach:

- $\rightsquigarrow$ construction of a parametrized class $p_{(\underline{\mathbf{y}}|\underline{\mathbf{x}};\underline{\mathbf{w}})}$ of (conditional) densities
- $\rightsquigarrow$ inference is based on good "generative models"

$$\underbrace{\boxed{\text{"generative" model of a data source}}}_{\substack{\text{statistical model of the} \\ \text{data generation process}}} \longrightarrow \boxed{\text{observations}}$$

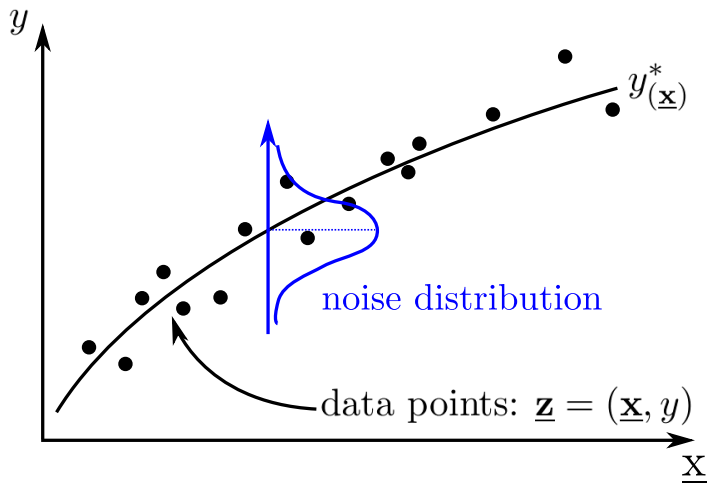
Comment

- The concept of generative models applies to supervised & unsupervised learning problems.
- Models $p_{(\underline{\mathbf{z}};\underline{\mathbf{w}})}$ for unconditional densities $\rightsquigarrow$ unsupervised learning (e.g., ICA, mixture models).
- Models $p_{(\underline{\mathbf{y}}|\underline{\mathbf{x}};\underline{\mathbf{w}})}$ for conditional densities $\rightsquigarrow$ supervised learning (e.g., "soft classification").

Example I: Generative models for regression

Statistical model of the data generation process:

$$y_{(\underline{\mathbf{x}})} = \underbrace{y^*_{(\underline{\mathbf{x}})}}_{\text{deterministic relationship}} + \underbrace{\eta}_{\text{zero-mean noise}}$$



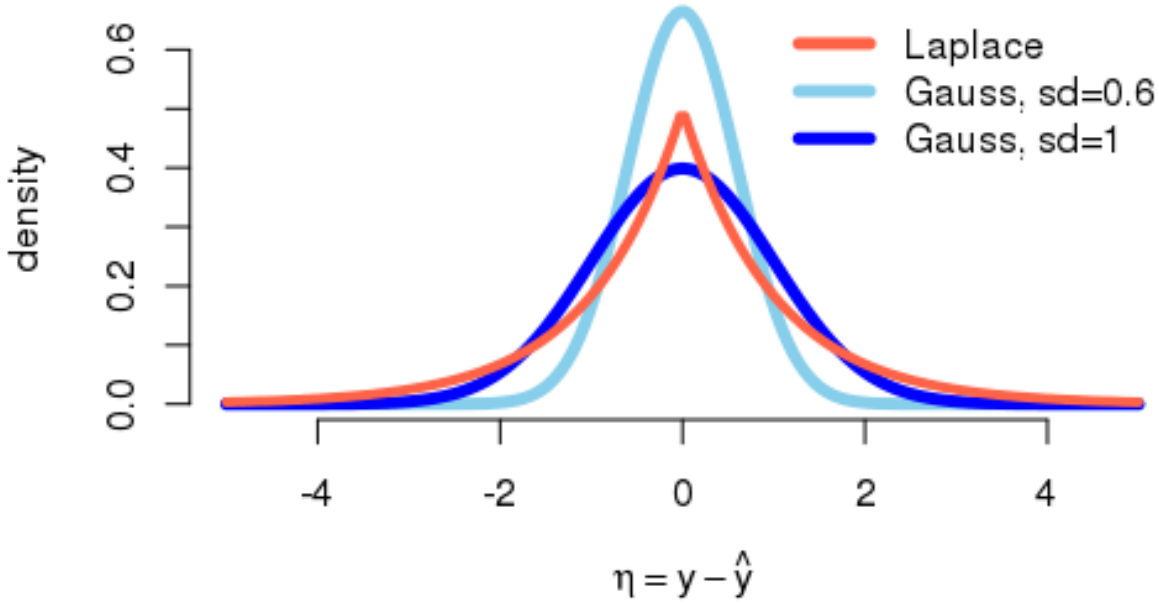
- Unknown deterministic relationship $y^*_{(\underline{\mathbf{x}})}$ approximated by a parametrized function $\hat{y}_{(\underline{\mathbf{x}};\underline{\mathbf{w}})}$ (e.g. an ANN).
- Unknown noise process η approximated by a parametrized distribution $\hat{p}(\eta;\sigma)$.
- Here: additive noise.
  - Other noise models possible (e.g. multiplicative noise).

Common noise models: Minkowski noise

- Noise distribution is given as:

$$\widehat{p}_{(y|\underline{\mathbf{x}};\underline{\mathbf{w}})} = \frac{d\beta^{\frac{1}{d}}}{2\,\underbrace{\Gamma(\frac{1}{d})}_{\text{Gamma function}}} \exp\Big\{ -\beta\big|y - \widehat{y}_{(\underline{\mathbf{x}};\underline{\mathbf{w}})}\big|^d \Big\}$$

- $d = 1$: Laplace distribution, $d = 2$: Gaussian distribution



Example II: Classification for M classes C_k

Description of the data generation process:

$$p_{(C_k|\underline{\mathbf{x}})}$$

$\rightsquigarrow$ Overlapping classes can induce 'label noise'.

Model:

$$\widehat{p}_{(C_k|\underline{\mathbf{x}};\underline{\mathbf{w}})} = y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}$$

$\rightsquigarrow$ parametrized function (e.g. an ANN)

3.3.2 Bayesian Model Selection

Degrees of belief

causal rules: cause → effect "generative model"

diagnostic rules: cause ← effect "inference"

generative model (M) / hypothesis

evidence (E) / observations

Degrees of belief

causal rules: cause → effect "generative model"

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generative model (M) / hypothesis

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Bayes rule

$$\underbrace{P_{(M|E)}}_{\text{posterior}} = \frac{\overbrace{P_{(E|M)}}^{\text{likelihood}}\,\overbrace{P_{(M)}}^{\text{prior}}}{\underbrace{P_{(E)}}_{\substack{\text{normalization constant}\\ \text{(“evidence”)}}}}$$

Likelihood and prior

$$P(M|E) = \frac{P(E|M)\,P(M)}{P(E)}$$

Likelihood $P(E|M)$**:** probability of observing the evidence E, given that model M is true $\Leftarrow$ generative model

Likelihood and prior

$$P(M|E) = \frac{P(E|M)\,P(M)}{P(E)}$$

Prior $P(M)$: maximum entropy methods $\rightarrow$ least informative prior belief

$$-\sum_m P_{(m)} \ln P_{(m)} \stackrel{!}{=} \max$$

Constraints on the prior

$$\sum_m P_{(m)} = 1; \qquad P_{(m)} \geq 0$$

uninformative prior: $P_{(m)} = \mathrm{const_m} \quad \rightsquigarrow \quad P_{(M|E)} \sim P_{(E|M)}$

Further constraints may follow from additional prior knowledge, e.g., about the parameter values of a parametrized model class (weights of an MLP). (cf. slides 18 & 19)

3.3.3 Bayesian Prediction

Bayesian committees

- fundamental problem of prediction

observations $E \longrightarrow$ degree of belief $P_{(e|E)}$ for a new event e

$$
\begin{aligned}
P_{(e|E)} &= \sum_m P_{(e,m|E)} && \text{marginalization} \\
&= \sum_m P_{(e|m,E)} P_{(m|E)} && \text{def. of conditional probability} \\
&\overset{!}{\approx} \sum_m P_{(e|m)} P_{(m|E)} && \text{conditional independence assumption}
\end{aligned}
$$

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\end{aligned}
$$

$$
\begin{tikzcd}
E \arrow[r] \arrow[rr, bend left] & M \arrow[r] & e
\end{tikzcd}
\quad \Rightarrow \quad
E \longrightarrow M \longrightarrow e
$$

- Bayesian committee: $P_{(e|E)} \approx \sum_m \underbrace{P_{(e|m)}}_{\text{likelihood}} \underbrace{P_{(m|E)}}_{\text{posterior}}$

Decision making: Minimizing expected loss

Cost of making a wrong prediction:

$$C(\underbrace{e}_{\text{true}}, \underbrace{\widehat{e}}_{\text{predicted}}) \qquad \text{e.g.} \quad |e - \hat{e}|^d$$

Other examples: 0-1 loss, robust error criterion.

- Decide for the value that minimizes the expected loss:

$$\widehat{e} \quad = \quad \operatorname*{argmin}_{\tilde{e}} \int C_{(e,\tilde{e})} P_{(e|E)} \, de$$

- Decision for the most probable value only if all errors are equally costly.

(see also Section 1.4.8)

3.3.4 Application: MLPs with Weight Decay

Recap: Bayes' theorem

$$\underbrace{P_{(M|E)}}_{\text{posterior}} = \frac{\overbrace{P_{(E|M)}}^{\text{likelihood}} \overbrace{P_{(M)}}^{\text{prior}}}{\underbrace{P_{(E)}}_{\substack{\text{normalization constant} \\ (\text{“evidence”})}}}$$

Construction of the data likelihood

training data: $\left\{ \left(\underline{\mathbf{x}}^{(\alpha)}, y_T^{(\alpha)} \right) \right\}$, $\alpha \in \{1, \ldots, p\}$,

abbreviations:

$X = \left\{ \underline{\mathbf{x}}^{(\alpha)} \right\}, Y = \left\{ \underline{\mathbf{y}}_T^{(\alpha)} \right\}$

Data likelihood

ansatz:

$$P_{\left(y_T^{(\alpha)} | \underline{\mathbf{x}}^{(\alpha)}; \underline{\mathbf{w}}\right)} \sim \exp\left(-\beta \, e_{\left(y_T^{(\alpha)}, \underline{\mathbf{x}}^{(\alpha)}; \underline{\mathbf{w}}\right)} \right) \quad e(\cdot) \text{ non-negative function}$$

Construction of the data likelihood

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$X = \left\{\underline{\mathbf{x}}^{(\alpha)}\right\}, Y = \left\{\underline{\mathbf{y}}_T^{(\alpha)}\right\}$

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assumption: training data drawn i.i.d. from the joint distribution

$$\begin{aligned} P_{(Y|X;\underline{\mathbf{w}})} &\sim \prod_\alpha \exp\left(-\beta\, e_{\left(y_T^{(\alpha)}, \underline{\mathbf{x}}^{(\alpha)}; \underline{\mathbf{w}}\right)}\right) \\ &\sim \exp\left(-\beta \sum_\alpha e_{\left(y_T^{(\alpha)}, \underline{\mathbf{x}}^{(\alpha)}; \underline{\mathbf{w}}\right)}\right) \\ &\sim \exp\left(-\beta\, E^T_{(Y,X;\underline{\mathbf{w}})}\right) \end{aligned}$$

Construction of the data likelihood

- Example: *additive Gaussian noise*

$$y_T^{(\alpha)} = \hat{y}_{(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})} + \hat{\eta} \qquad \text{with} \quad \hat{\eta} \sim \mathcal{N}(0, \sigma^2)$$

$$P_{(Y|X;\underline{\mathbf{w}})} = \frac{1}{(2\pi\sigma^2)^{\frac{p}{2}}} \exp\left\{ -\frac{1}{2\sigma^2} \underbrace{\sum_{\alpha=1}^{p}}_{\substack{\text{iid}\\ \text{assumption}}} \left(y_T^{(\alpha)} - \underbrace{\widehat{y}_{(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})}}_{\rightarrow \text{ MLP}} \right)^2 \right\}$$

Construction of the data likelihood

- Example: *additive Gaussian noise*

$$y_T^{(\alpha)} = \hat{y}_{(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})} + \hat{\eta} \qquad \text{with} \quad \hat{\eta} \sim \mathcal{N}(0, \sigma^2)$$

$$\begin{aligned}
P_{(Y|X;\underline{\mathbf{w}})} &= \underbrace{\frac{1}{(2\pi\sigma^2)^{\frac{p}{2}}}}_{\frac{1}{Z}} \exp\Bigg\{ - \underbrace{\frac{1}{\sigma^2}}_{\beta} \overbrace{\sum_{\alpha=1}^{p} \underbrace{\tfrac{1}{2}\Big(y_T^{(\alpha)} - \widehat{y}_{(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})}\Big)^2}_{\substack{\text{individual loss} \\ e(y_T^{(\alpha)},\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})}}}^{\text{training error } E^T\text{: quadratic error}^1} \Bigg\} \\
&= \frac{1}{Z} \exp\Big\{ - \beta \sum_{\alpha=1}^{p} e(y_T^{(\alpha)}, \underline{\mathbf{x}}^{(\alpha)}; \underline{\mathbf{w}}) \Big\} \\
&= \frac{1}{Z} \exp\Big(- \beta\, E^T_{(Y,X;\underline{\mathbf{w}})} \Big)
\end{aligned}$$

- maximizing the likelihood $P_{(Y|X;\underline{\mathbf{w}})}$ $\sim$ minimizing the quadratic error E^T

[1] Here, E^T is defined as the sum rather than the average of the individual losses, hence E^T scales linearly with p.

Choice of the prior

- **Goal:** Find the most "unprejudiced" distribution consistent with our prior knowledge ("constraints").

Ansatz: The maximum entropy method

$$
\begin{aligned}
-\sum_{\underline{\mathbf{w}}} P_{(\underline{\mathbf{w}})} \ln P_{(\underline{\mathbf{w}})} &\stackrel{!}{=} \max \\
\sum_{\underline{\mathbf{w}}} P_{(\underline{\mathbf{w}})} &= 1 \qquad \text{(normalization)} \\
\sum_{\underline{\mathbf{w}}} E^R_{(\underline{\mathbf{w}})} P_{(\underline{\mathbf{w}})} &= E_0 \qquad \text{(prior knowledge)}
\end{aligned}
$$

Choice of the prior

Solution using Lagrange multipliers

$$-\sum_{\underline{\mathbf{w}}} P_{(\underline{\mathbf{w}})} \ln P_{(\underline{\mathbf{w}})} + \lambda \left(\sum_{\underline{\mathbf{w}}} P_{(\underline{\mathbf{w}})} - 1 \right) - \alpha \left(\sum_{\underline{\mathbf{w}}} E^R_{(\underline{\mathbf{w}})} P_{(\underline{\mathbf{w}})} - E_0 \right) \stackrel{!}{=} \max$$

Choice of the prior

Solution using Lagrange multipliers

$$-\sum_{\underline{\mathbf{w}}} P_{(\underline{\mathbf{w}})} \ln P_{(\underline{\mathbf{w}})} + \lambda\left(\sum_{\underline{\mathbf{w}}} P_{(\underline{\mathbf{w}})} - 1\right) - \alpha\left(\sum_{\underline{\mathbf{w}}} E^R_{(\underline{\mathbf{w}})} P_{(\underline{\mathbf{w}})} - E_0\right) \stackrel{!}{=} \max$$

$$\begin{aligned} -\ln P_{(\underline{\mathbf{w}})} - 1 + \lambda - \alpha E^R_{(\underline{\mathbf{w}})} &= 0 \\ \ln P_{(\underline{\mathbf{w}})} &= \lambda - 1 - \alpha E^R_{(\underline{\mathbf{w}})} \\ P_{(\underline{\mathbf{w}})} &\sim \exp\left(-\alpha E^R_{(\underline{\mathbf{w}})}\right) \end{aligned}$$

- λ is found through normalization of prior probabilities
  $\rightarrow$ equivalent to choosing a normalization factor.
- α can be calculated - in principle - from the corresponding constraint, however, it is often used as a hyperparameter itself.

Computing the posterior degree of belief

- Bayes rule:

$$P_{(\underline{\mathbf{w}}|Y,X)} \sim P_{(Y|X;\underline{\mathbf{w}})} P_{(\underline{\mathbf{w}})}$$

$$\sim \exp\{-\tfrac{1}{\sigma^2} E^T - \alpha E^R\} \quad = \quad \exp(-\tfrac{1}{\sigma^2} R)$$

- The model which we believe in most strongly is the model which minimizes the "cost function" R.
- The more data points, the less important the prior becomes:

$$R = \underbrace{E^T}_{\sim\#\text{data}} + \underbrace{\alpha' E^R}_{\sim\#\text{parameters}} \quad \text{with } \alpha' = \alpha\sigma^2$$

- Examples: weight decay, $E^R_{(\underline{\mathbf{w}})} = \frac{1}{2}\sum_i w_i^2$, or Lasso, $E^R_{(\underline{\mathbf{w}})} = \sum_i |w_i|$

.

Example: Additive Gaussian noise and weight decay

$$P(\underline{\mathbf{w}}|Y,X) \quad \sim \quad \exp\left(-\frac{1}{\sigma^2}R\right)$$

$$\text{with} \qquad R \quad = \quad \frac{1}{2}\sum_{\alpha=1}^{p}\left(y_T^{(\alpha)} - \widehat{y}_{(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})}\right)^2 \quad + \quad \frac{\alpha'}{2}\sum_{k=1}^{d}\mathrm{w}_k^2$$

- Maximizing the posterior $P(\underline{\mathbf{w}}|Y,X)$
  $\rightsquigarrow$ minimizing the regularized training error R.

Recap: Section 1.4.6

$$R_{[\underline{\mathbf{w}}]} = \underbrace{E^T_{[\underline{\mathbf{w}}]}}_{\substack{\text{training}\\ \text{error}}} + \underbrace{\lambda E^R_{[\underline{\mathbf{w}}]}}_{\substack{\text{regularization}\\ \text{term}}} \stackrel{!}{=} \min$$

E^R : penalizes certain models $\rightsquigarrow$ "soft" restrictions on model space

λ : regularization parameter: trade-off between observations and prior knowledge

3.3.5 The "maximum a posteriori" Method

Prediction by Bayesian committee

$$P_{(y|\underline{\mathbf{x}};Y,X)} = \int P_{(y|\underline{\mathbf{x}};\underline{\mathbf{w}})} \, P_{(\underline{\mathbf{w}}|Y,X)} \, d\underline{\mathbf{w}}$$

(see Bishop, Chapter 5.7)

Prediction by Bayesian committee

$$P_{(y|\underline{\mathbf{x}};Y,X)} = \int P_{(y|\underline{\mathbf{x}};\underline{\mathbf{w}})}\, P_{(\underline{\mathbf{w}}|Y,X)}\, d\underline{\mathbf{w}}$$

$$\begin{aligned} P_{(y|\underline{\mathbf{x}};\underline{\mathbf{w}})} &= \frac{1}{\sigma\sqrt{2\pi}} \exp\Big(-\frac{1}{\sigma^2} e_{(y,\underline{\mathbf{x}};\underline{\mathbf{w}})} \Big) \\ P_{(\underline{\mathbf{w}}|Y,X)} &= \frac{1}{(2\pi\sigma^2)^{p/2}} \exp\Big(-\frac{1}{\sigma^2} \sum_{\alpha=1}^{p} e_{(y_T^{(\alpha)},\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})} - \alpha E^R_{[\underline{\mathbf{w}}]} \Big) \end{aligned}$$

(see Bishop, Chapter 5.7)

Prediction by Bayesian committee

$$P_{(y|\underline{\mathbf{x}};Y,X)} = \int P_{(y|\underline{\mathbf{x}};\underline{\mathbf{w}})}\, P_{(\underline{\mathbf{w}}|Y,X)}\, d\underline{\mathbf{w}}$$

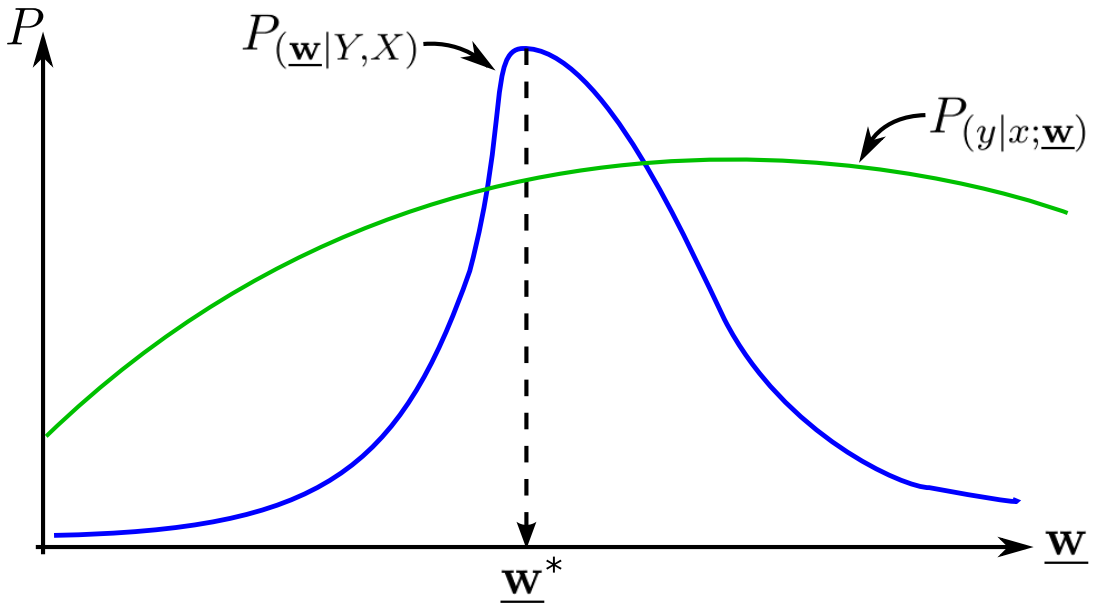
- There is no closed expression for the integral for many models.
- Numerical solutions, e.g., using MCMC methods.
- For some cases, the integral can be evaluated analytically, e.g. for:
  - regression with quadratic cost $e_{(y,\underline{\mathbf{x}};\underline{\mathbf{w}})} = \frac{1}{2}(y - \hat{y}_{(\underline{\mathbf{x}};\underline{\mathbf{w}})})^2$,
  - using linear functions $\hat{y}_{(\underline{\mathbf{x}};\underline{\mathbf{w}})} = \underline{\mathbf{w}}^\top \underline{\mathbf{x}}$,
  - and weight decay regularization $E^R_{[\underline{\mathbf{w}}]} = \frac{1}{2}\underline{\mathbf{w}}^\top \underline{\mathbf{w}}$.
- Exact evaluation of the integral, but using approximations for the integrand (e.g., via variational methods).

(see Bishop, Chapter 5.7)

The maximum a posteriori approximation (MAP)

- **Assumption:** Posterior has a localized maximum.

$$P_{(y|\underline{\mathbf{x}};Y,X)} = \int P_{(y|\underline{\mathbf{x}};\underline{\mathbf{w}})}\, P_{(\underline{\mathbf{w}}|Y,X)}\, d\underline{\mathbf{w}}$$



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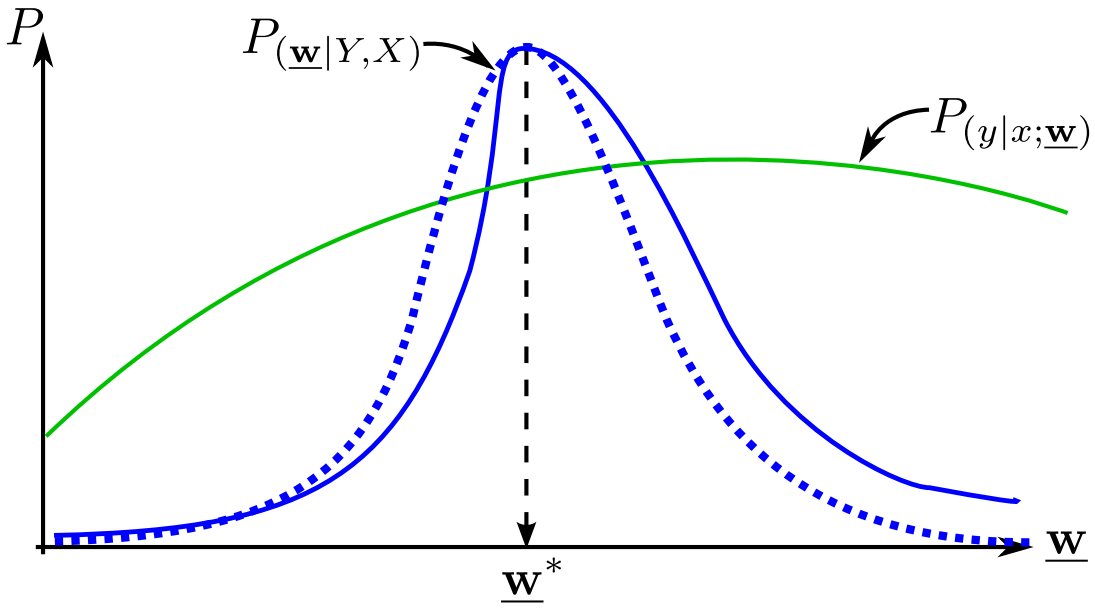


The maximum a posteriori approximation (MAP)

$$P_{(y|\underline{\mathbf{x}};Y,X)} \sim \int \underbrace{\exp\left(-\frac{\beta}{2}e_{(y,x;\underline{\mathbf{w}})}\right)}_{\substack{\text{generative model} \\ \text{/ likelihood}}} \underbrace{\exp\left(-\frac{\beta}{2}R_{(Y,X;\underline{\mathbf{w}})}\right)}_{\text{posterior}} d\underline{\mathbf{w}}$$

1. Gauss-approximation of posterior
   - Taylor expansion up to second order[2] around the maximum of its exponent at $\underline{\mathbf{w}}^*$

$$R_{(Y,X;\underline{\mathbf{w}})} = R_{(Y,X;\underline{\mathbf{w}}^*)} + \frac{1}{2}\sum_{i,j}(\mathrm{w}_i - \mathrm{w}_i^*) \underbrace{\frac{\partial^2 R}{\partial \mathrm{w}_i \partial \mathrm{w}_j}}_{H_{ij}\text{: Hessian}} \Bigg|_{\underline{\mathbf{w}}^*} (\mathrm{w}_j - \mathrm{w}_j^*)$$

2. Linear approximation of the exponent of the individual likelihood
   - Taylor expansion up to first order around $\underline{\mathbf{w}}^*$

$$e_{(y,\underline{\mathbf{x}};\underline{\mathbf{w}})} = e_{(y,\underline{\mathbf{x}};\underline{\mathbf{w}}^*)} + \sum_i \frac{\partial e}{\partial \mathrm{w}_i}\Bigg|_{\underline{\mathbf{w}}^*} (\mathrm{w}_i - \mathrm{w}_i^*)$$

[2] First order terms vanish, because $\underline{\mathbf{w}}^*$ is the location of the maximum.

The predictive distribution

- The MAP approximation yields a closed form solution for the predictive distribution

$$P_{(y|\underline{\mathbf{x}};Y,X)} \sim \exp\left(-\beta e + \frac{\beta}{2}\left(\frac{\partial e}{\partial \underline{\mathbf{w}}}\right)^{\top} \underline{\mathbf{H}}^{-1} \frac{\partial e}{\partial \underline{\mathbf{w}}}\right)\Bigg|_{\underline{\mathbf{w}}^*}$$

(*calculation see supplementary material*)

Example: MLP with weight decay

$$P_{(y|\underline{\mathbf{x}};Y,X)} \sim \exp\left(-\beta e + \frac{\beta}{2}\left(\frac{\partial e}{\partial \underline{\mathbf{w}}}\right)^{\top} \underline{\mathbf{H}}^{-1} \frac{\partial e}{\partial \underline{\mathbf{w}}}\right)\Bigg|_{\underline{\mathbf{w}}^*}$$

- example assumptions:

$$\beta = \frac{1}{\sigma^2}, \quad e_{(y,\underline{\mathbf{x}};\underline{\mathbf{w}}^*)} = \frac{1}{2}\big(y - \hat{y}_{(\underline{\mathbf{x}};\underline{\mathbf{w}}^*)}\big)^2, \quad \frac{\partial e}{\partial \underline{\mathbf{w}}}\Big|_{\underline{\mathbf{w}}^*} = -\big(y - \widehat{y}_{(\underline{\mathbf{x}};\underline{\mathbf{w}})}\big) \underbrace{\frac{\partial \widehat{y}}{\partial \underline{\mathbf{w}}}\Big|_{\underline{\mathbf{w}}^*}}_{\underline{\mathbf{g}}}$$

- approximate predictive distribution:

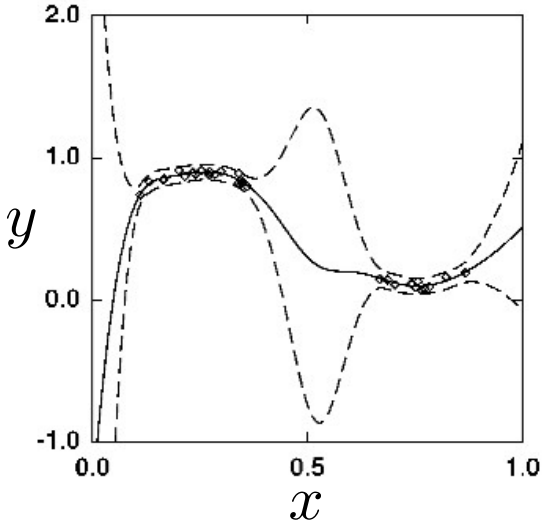
$$P_{(y|\underline{\mathbf{x}};Y,X)} \sim \exp\left\{ -\frac{1}{\sigma^2}\big(1 - \underline{\mathbf{g}}^{\top}\underline{\mathbf{H}}^{-1}\underline{\mathbf{g}}\big) \cdot \ \frac{1}{2}\big(y - \widehat{y}_{(\underline{\mathbf{x}};\underline{\mathbf{w}}^*)}\big)^2 \right\}$$

Example: MLP with weight decay

$$P_{(y|\underline{\mathbf{x}};Y,X)} \sim \exp\left\{ -\underbrace{\frac{1}{\sigma^2}\left(1 - \underline{\mathbf{g}}^\top \underline{\mathbf{H}}^{-1} \underline{\mathbf{g}}\right)}_{1/\sigma_y^2} \cdot \tfrac{1}{2}\left(y - \widehat{y}_{(\underline{\mathbf{x}};\underline{\mathbf{w}}^*)}\right)^2 \right\}$$

- predictive distribution is Gaussian with mean $\widehat{y}_{(\underline{\mathbf{x}};\underline{\mathbf{w}}^*)}$ and

$$\sigma_y^2 \quad \stackrel{!}{=} \quad \frac{\sigma^2}{\underbrace{1}_{\substack{\text{noise}\\ \text{model}}} - \underbrace{\underline{\mathbf{g}}^\top \underline{\mathbf{H}}^{-1} \underline{\mathbf{g}}\Big|_{\underline{\mathbf{w}}^*}}_{\substack{\text{correction for}\\ \text{parameter uncertainty}}}} \qquad \text{(predictive variance)}$$



Comments

1. $\underline{\mathbf{w}}^*$ is referred to as the "MAP-solution":

$$\underline{\mathbf{w}}^* = \underset{\underline{\mathbf{w}}}{\operatorname{argmin}} \left(E^T + \alpha' E^R\right)$$

   Formal equivalence between MAP solution and regularized ERM:

$$E^T \widehat{=} -\log \text{likelihood} \qquad\qquad E^R \widehat{=} -\log \text{prior}$$

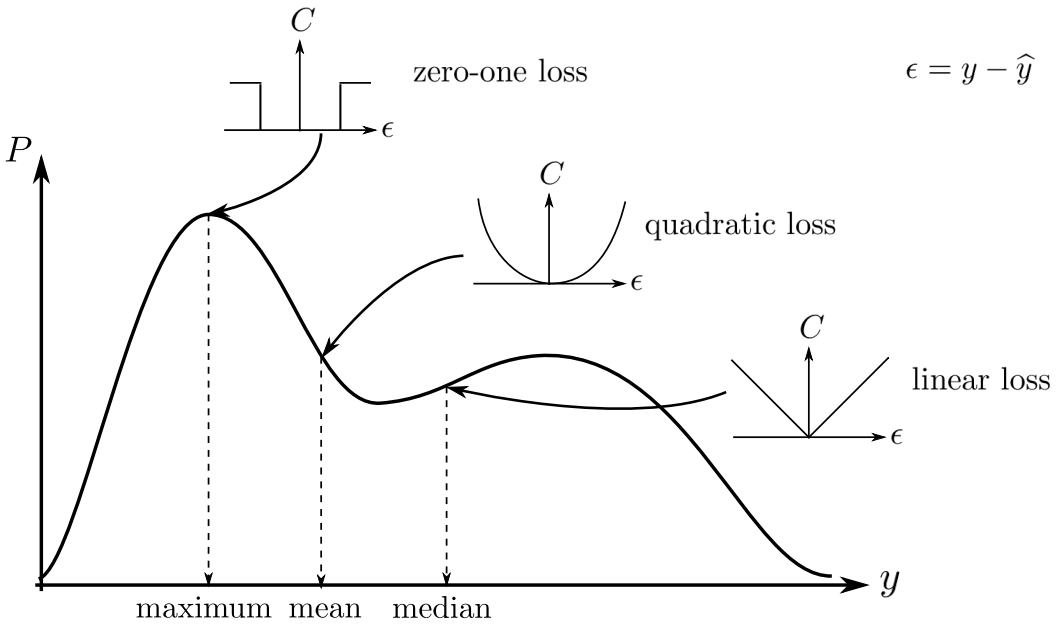
2. For MLPs, both $\underline{\mathbf{g}}$ and $\underline{\mathbf{H}}^{-1}$, can be calculated efficiently, cf. Bishop (2006).

3. The MAP solution accounts for two types of uncertainty:
   - uncertainty inherent in the data generation process (σ^2),
   - precision of the estimated model (via $\underline{\mathbf{H}}^{-1}$).

Application: point prediction of attributes

- Find the prediction $\hat{y}$ that minimizes the *expected loss*
  - for a given cost function $C_{(y,\tilde{y})}$
  - and for the predictive distribution $P_{(y|\underline{\mathbf{x}};Y,X)}$.

$$\widehat{y}_{(\underline{\mathbf{x}})} = \operatorname*{argmin}_{\tilde{y}} \int dy\, C_{(y,\tilde{y})}\, P_{(y|\underline{\mathbf{x}};Y,X)}$$



- Gaussian distribution (e.g., MAP approximation): maximum $\widehat{=}$ mean $\widehat{=}$ median

References I

C.M. Bishop. *Pattern Recognition and Machine Learning*. Springer Verlag, 2006. ISBN 978-0-387-31073-2.